MIL-C-18757/3(OS)
8 August 1974
SUPERSEDING
MIL-C-18859(NOrd)
29 June 1955
MIL-C-18859
AMENDMENT 1
2 October 1959
and
WS 13512
18 August 1970

MILITARY SPECIFICATION

CARTRIDGE CASE, STEEL, 5-INCH, MARK 10

ALL MODS (38 CALIBER)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the specific requirements for the 5"/38 caliber Mk 10 all Mods Steel Cartridge Case, hereinafter referred to as the "cartridge case". The general requirements are covered in MIL-C-18757.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-Z-325	Zinc Coating, Electrodeposited, Requirements For

Military

MIL-C-18757	Cartridge Cases, Steel (Other Than Small Arms) General Specification for

DRAWINGS

Naval Ordnance Systems Command (Code Ident 10001)

517537	5-Inch Cartridge Case Mk 10 Mod 0 (38 Caliber) Steel
2847227	Case, Cartridge Mk 10 Mod 1 5"/38 Caliber, Steel

FSC 1320

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2.2 Other Publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

IHSP 73-100 Indian Head Special Publications, "Black Speck Rating Procedure for Cartridge Cases"

(Application for copies should be addressed to Naval Ordnance Station, Indian Head, Maryland 20640).

3. REQUIREMENTS

3.1 General Requirements. The complete requirements for procuring the cartridge case described herein shall consist of specification MIL-C-18757 and this document.

3.2 Preproduction Lot. The preproduction lot shall meet all the requirements specified herein and on NAVORD Drawing 517537 or 2847227, as applicable.

3.3 Production Lot. The production lot shall meet all the requirements specified herein and on NAVORD Drawing 517537 or 2847227, as applicable.

3.4 Hardness. The hardness values shall be as specified in Table I in the regions shown by Figure 1.

TABLE I

HARDNESS VALUES[1]

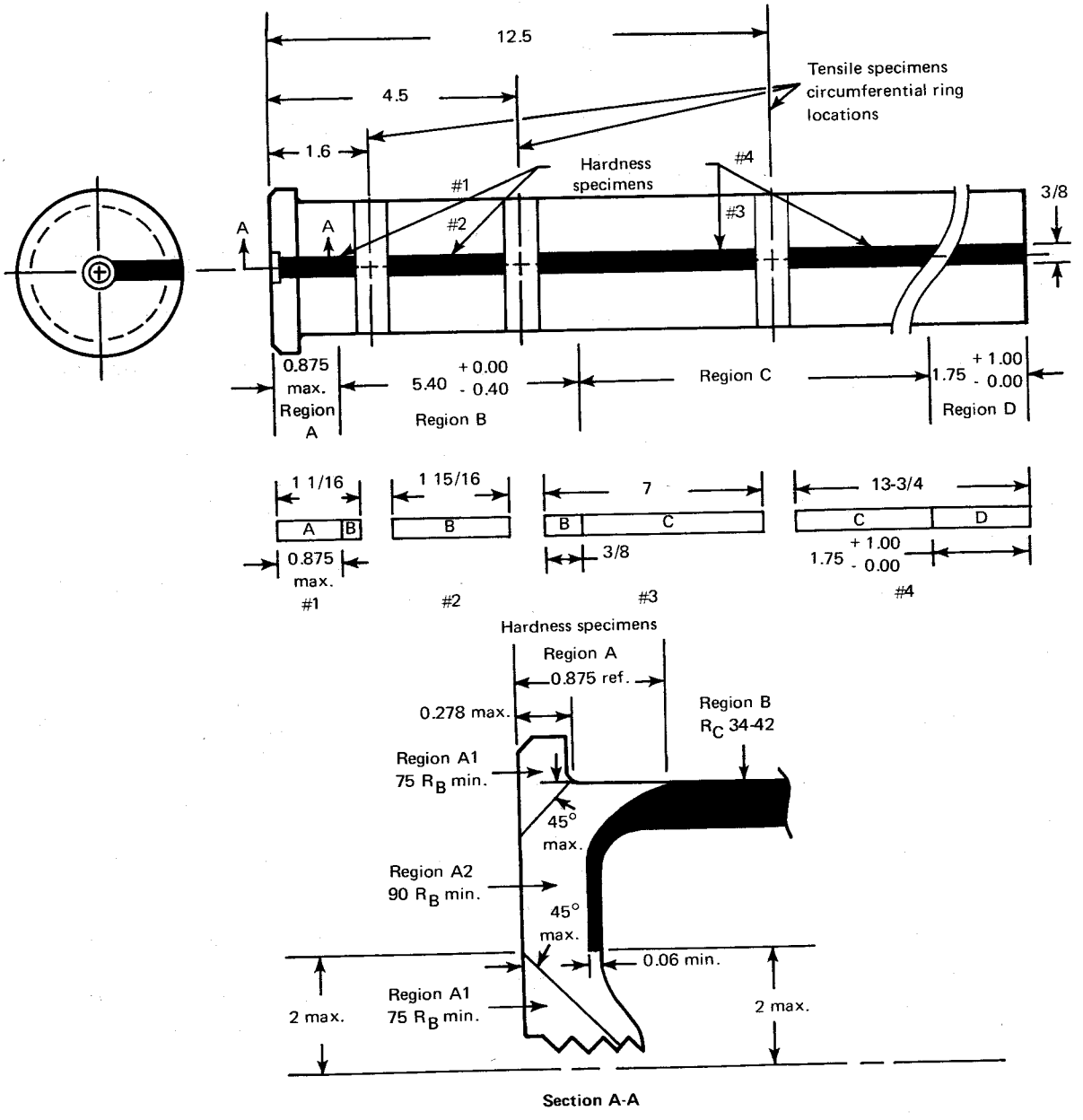
Region A_1		Region A_2		Region B		Region C		Region D
Max[2]	Min	Max[2]	Min	Max[2]	Min	Max[2]	Min	Max
42C	75B	42C	90B	42C	34C	42C	95B	85B

[1]All values given in Rockwell Scale.

[2]Paragraph 3.3.5 of QQ-Z-325B does not apply.

3.4.1 Nonconformance. When any one sample from Region B has hardness values outside the range specified in Table I, tensile tests, in addition to the normal tension requirements, shall be required. Centerline of specimens for the additional tensile tests shall be taken as follows: two samples at 2.0 inches from the base and two samples at 5.25 inches from the base. Samples at each location shall be taken 180 degrees apart. Additional tensile specimens shall meet the requirements of 3.5.

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Section A-A

Tolerances: Fractions, ± 1/32
One place decimals, ± 0.1

Dimensions: Unless otherwise specified, all dimensions are in inches

FIGURE 1. HARDNESS/TENSILE SPECIMEN LOCATIONS

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3.5 Tension. The yield strength and elongation values shall be as specified in Table II at the locations shown by Figure 1.

TABLE II

YIELD STRENGTH AND ELONGATION

Region	Yield Strength (Offset=0.2 percent)	Elongation in 1"
B	135,000 psi	6.0 percent
C	90,000 psi	5.0 percent

3.6 Protective Coating. The cartridge case protective coating shall have a rating of the black corrosion products of zinc less than the established standard of 0.24000 when tested in accordance with 4.4.3.

3.7 Ballistics. Each lot of cartridge cases shall pass the ballistic test specified in 4.4.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions for the cartridge case described herein shall consist of specification MIL-C-18757 and this document.

4.2 Preproduction Lot. The preproduction lot shall consist of 158 cartridge cases and shall be examined and tested in accordance with the Preproduction and Quality Conformance Inspection and Sampling Table of MIL-C-18757.

4.3 Production Lot. Unless otherwise specified in the contract or purchase order, the production lot shall consist of 5,000 cartridge cases plus those required for test purposes and shall be examined and tested in accordance with the Preproduction and Quality Conformance Inspection and Sampling Table of MIL-C-18757.

4.4 Tests.

4.4.1 Hardness. Cartridge cases shall be prepared and tested in accordance with the Hardness Test of MIL-C-18757.

4.4.2 Tension. Cartridge cases shall be prepared and tested in accordance with the Tension Test of MIL-C-18757.

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4.4.3 Protective Coating. Cartridge cases shall be prepared and tested in accordance with the Protective Coating Test of MIL-C-18757. The assigned plating thickness value of the interior recessed area of the cartridge case head shall be determined as follows:

Three specimen sections shall be taken from each case, approximately 120° apart, and six measurements, approximately 0.1 inch apart, shall be taken on each segment (see Figure 2).

The assigned plating thickness value (T) shall be calculated as follows:

T = plating thickness, any case
t_a = plating thickness, segment a
t_b = plating thickness, segment b
t_c = plating thickness, segment c
t_{a1} = plating thickness reading preceding t_{a2}
t_{a2} = minimum plating thickness reading of segment a
t_{a3} = plating thickness reading following t_{a2}

$$t_a = \frac{t_{a1} + 6\,t_{a2} + t_{a3}}{8}$$

$$T = \frac{t_a + t_b + t_c}{3}$$

t_b and t_c are computed in a similar manner

NOTE: (1) If t_{a2} is a measurement at location 1 or 6, use $t_{a1} = t_{a3}$ in above formula (use on any segment). See figure 2.

(2) Record all thickness readings.

The black speck corrosion rating procedure shall be as specified in IHSP 73-100.

4.4.4 Ballistics. Cartridge cases from each lot shall be tested in the quantities and for the defects as listed in Table III. Proof charge rounds shall have a test pressure between 19.0 and 21.0 TSI copper and all shall be fired in a single-shot mode. Service charge rounds shall have a test pressure between 17.0 and 19.0 TSI copper. For service pressure test, fire five rounds in single shot action with copper crusher gages to determine compliance with the pressure requirements and fire the remainder of the rounds in semi-automatic rapid fire action. Temperature of all rounds shall be as specified in Table III.

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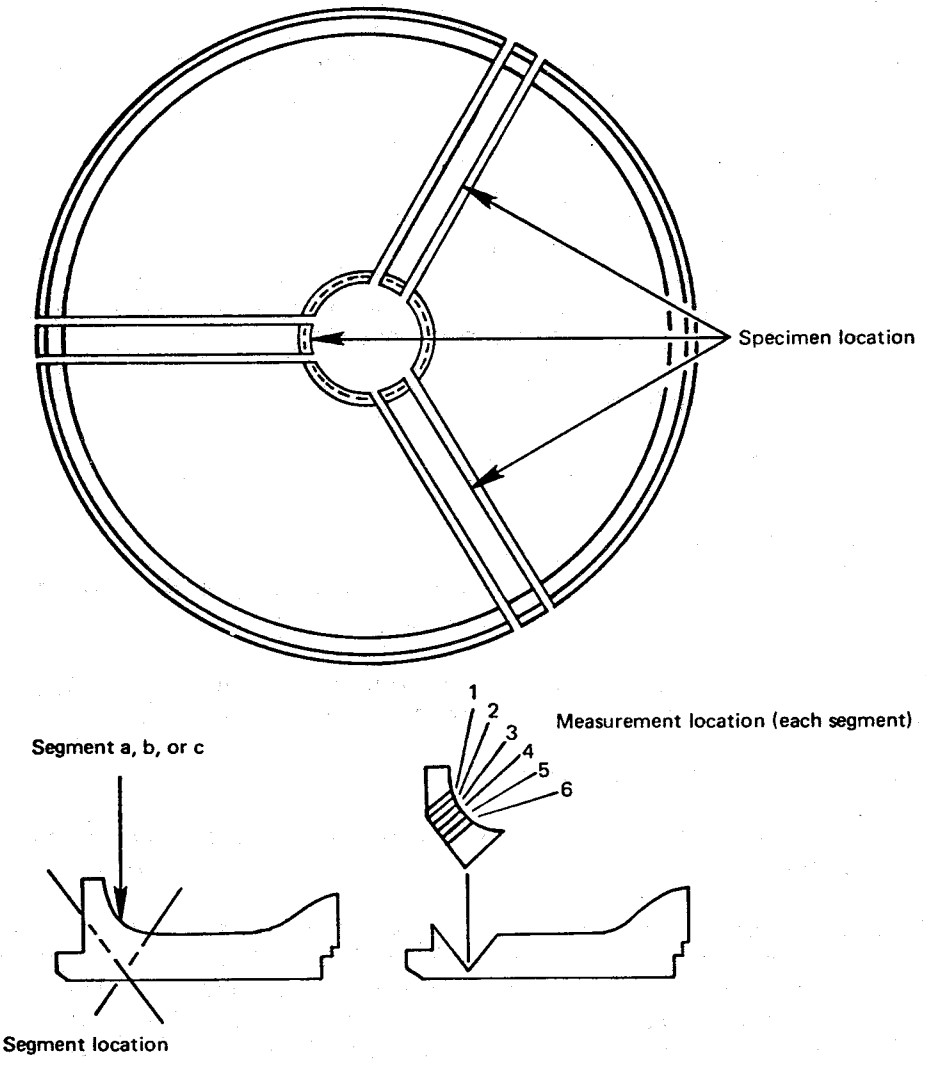


FIGURE 2. METALLOGRAPHIC SECTIONING

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TABLE III

BALLISTIC TEST CRITERIA

Type of Lot	No. of Rounds	Type of Rounds	Round Temp.	Class of Defect[1] (a) Acc	(a) Rej	(b) Acc	(b) Rej	(c) Acc	(c) Rej
Preproduction	70	Service	90°F	0	1	1	2	5	6
	30	Proof	90°F	0	1	1	2	5	6
	25	Service	120°F	0	1	1	2	5	6
	25	Service	-30°F	0	1	1	2	5	6
Production	40	Service	90°F	0	1	0[2]	2	2	3
	10	Proof	90°F	0	1	0[2]	2	2	3
	80	Service	90°F	0	1	0	1		
	20	Proof	90°F	0	1	0	1		

1. Class (a) Defect - Splits or fractures in the head or sidewall within a distance of three inches from the base; Class (b) Defect - failure of case to eject under automatic action or to extract in single-shot mode during normal ejection or extraction operation; Class (c) Defect - Splits or fractures other than class (a).

2. Reject on two or more defects. Test an additional 100 cases if one defect occurs. No additional class (a) or (b) defects are allowed.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with specification MIL-C-18757 and as specified herein.

5.2 Packing. Packing shall be in accordance with MIL-C-18757 and the following:

(a) Four (4) cartridge cases shall be in each container.

(b) Box dimensions shall be 12 11/16 x 12 11/16 x 26 7/8 inches.

(c) Partition dimensions shall be 31 5/8 x 12 5/8 inches, two required per container, one center slotted 19 inches and one center slotted 12 11/16 inches for assembly, two fold-out flaps 6 5/16 inches per partition for bottom pad.

(d) Top pad dimensions shall be 15 9/16 x 12 9/16 inches with 1 1/2 inch flange with four holes 5 7/16 inches in diameter.

Note: Tolerances for all packing dimensions shall be ±1/32 inch.

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6. NOTES

6.1 Intended Use. The 5"/38 caliber Mk 10, all mods, cartridge cases, when assembled with their appropriate primers, propellant charges, and plugs, and when used with their appropriate loaded projectiles, form complete rounds which are used as service ammunition in the 5"/38 caliber gun.

6.2 Ordering Data. Ordering data shall be in accordance with specification MIL-C-18757.

Custodian:
Navy - OS

Preparing Activity:
Navy - OS
(Project No. 1320-N237-3)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-18757/3(OS) Cartridge Case, Steel, 5-Inch, Mark 10 All Mods (38 Caliber)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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